

Robot Control I: Reactive Control

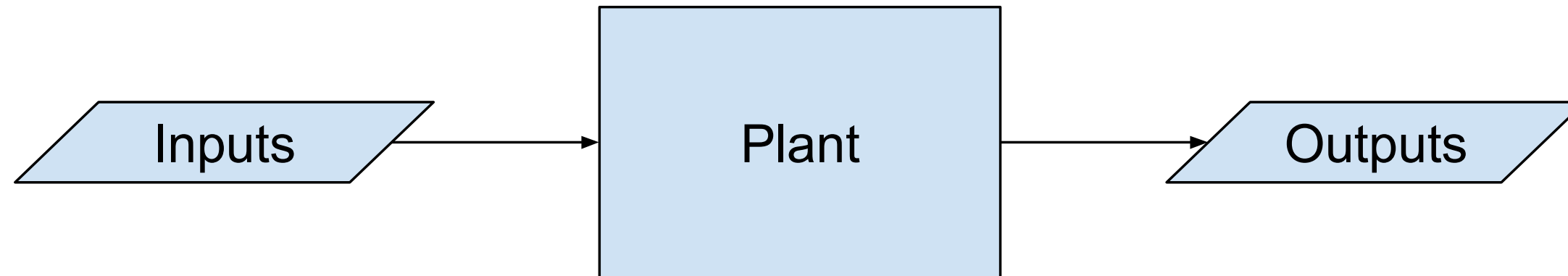
MEGN 441
Zhaoda Du

Basics of Robotic Control

Three primitives of robotics

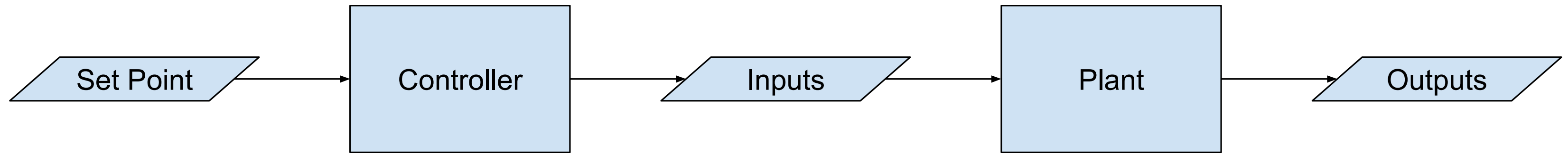
- Sense (Sensors)
 - Provides information from the sensors as measurements and other data describing the world and robot's state for the other systems
- Plan (Controllers)
 - Reads the information from sensing, combines with current world knowledge (models), and determines the actions to take
- Act (Actuators)
 - Send the physical commands to the system and actuators to generate the plant output

Open Loop control



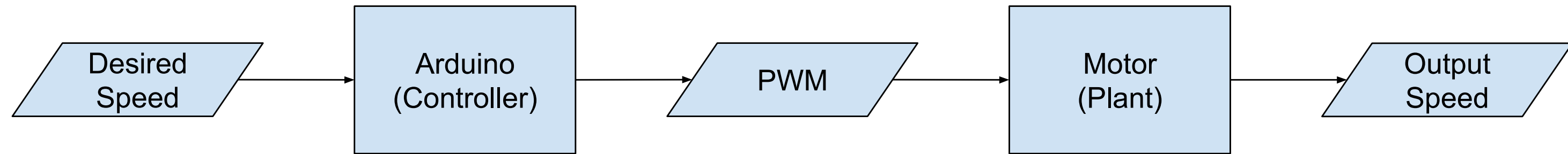
- Plant is the model used to represent a physical system
- Inputs are independent variables the operator can change
- Outputs are dependent variables produced by the systems reaction/response

Open Loop Control



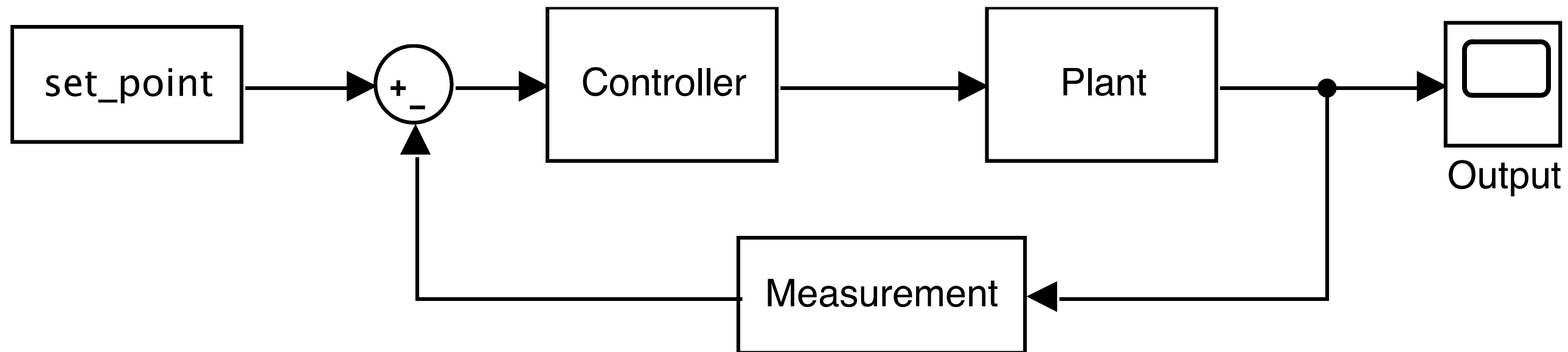
- Controller changes the plant inputs
- No measurement from the output
- No correction for errors

Open Loop Control



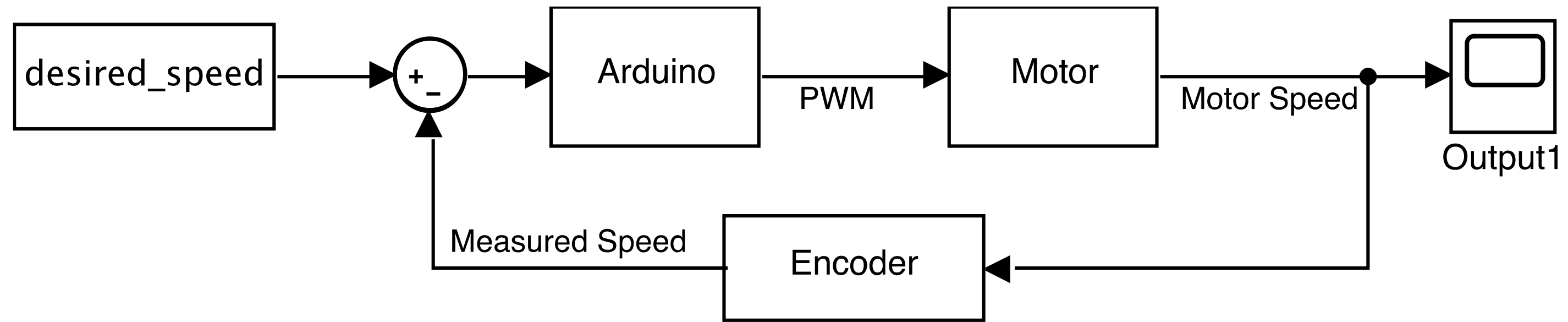
- Your lab 2 robots

Closed Loop Control



- Incorporates feedback signals
- Self-regulates based on output error or sensor measurements
- Controller must be tuned to keep the system stable

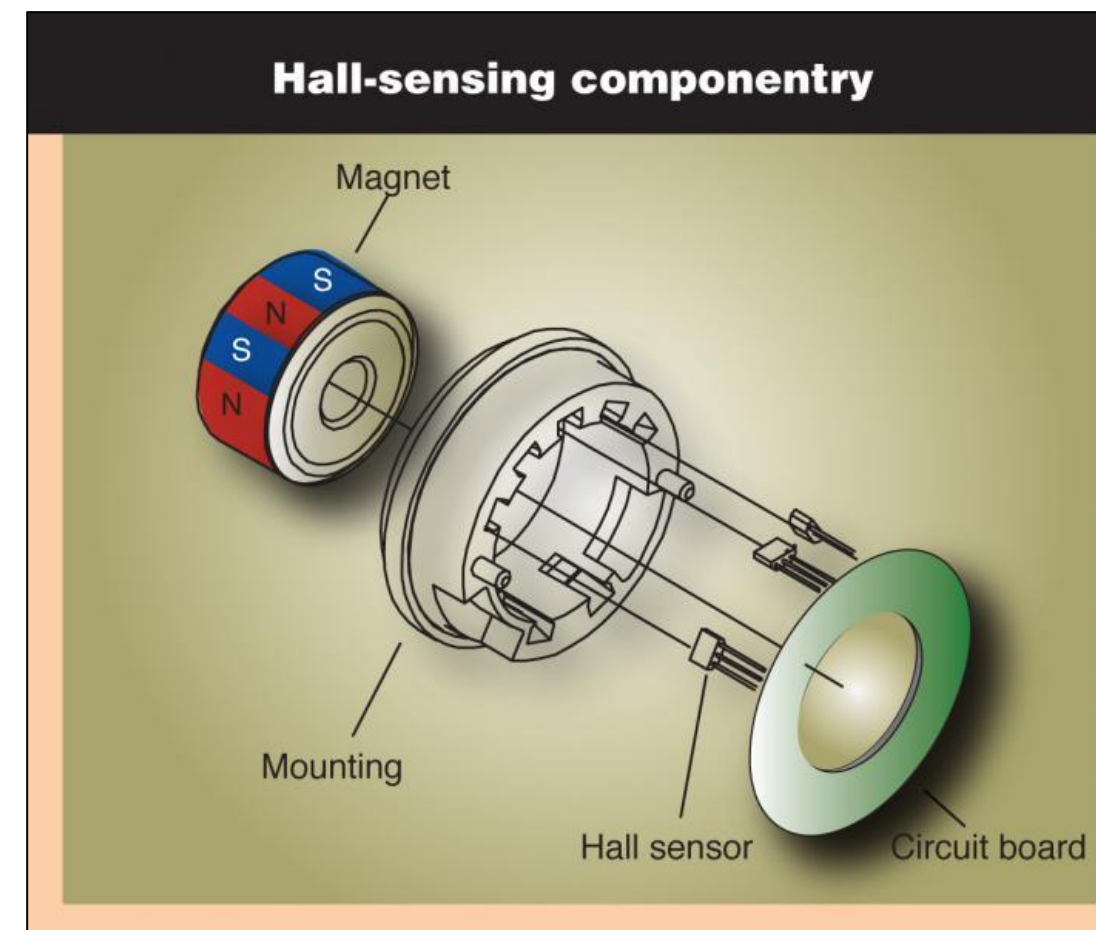
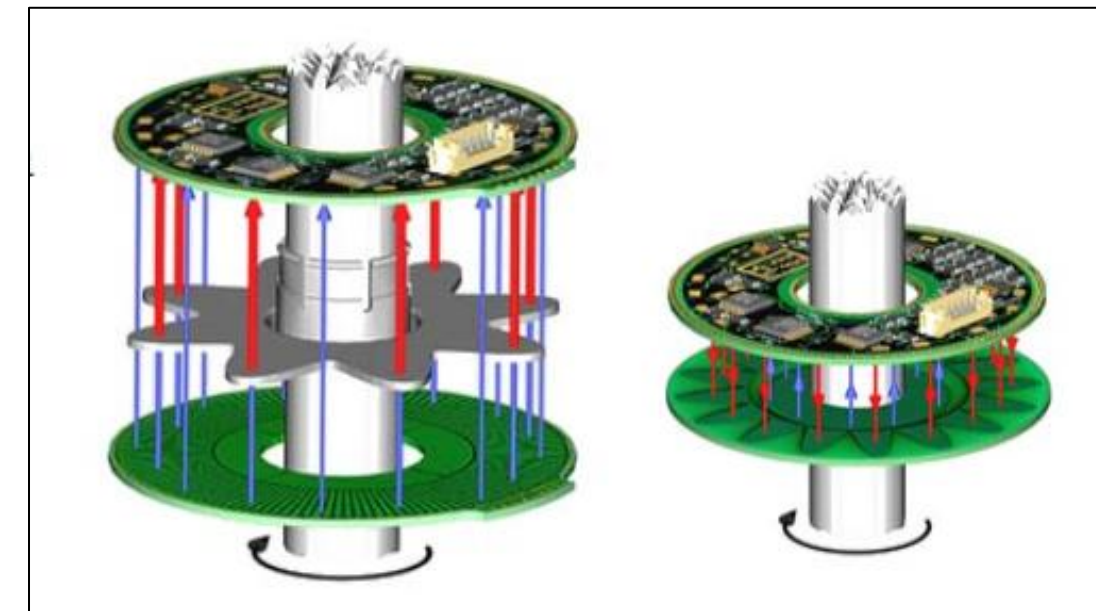
Closed Loop Control



- Requires an encoder to read the motor's output speed
- >10Hz is technically considered real-time
- >30Hz is more likely to behave real-time

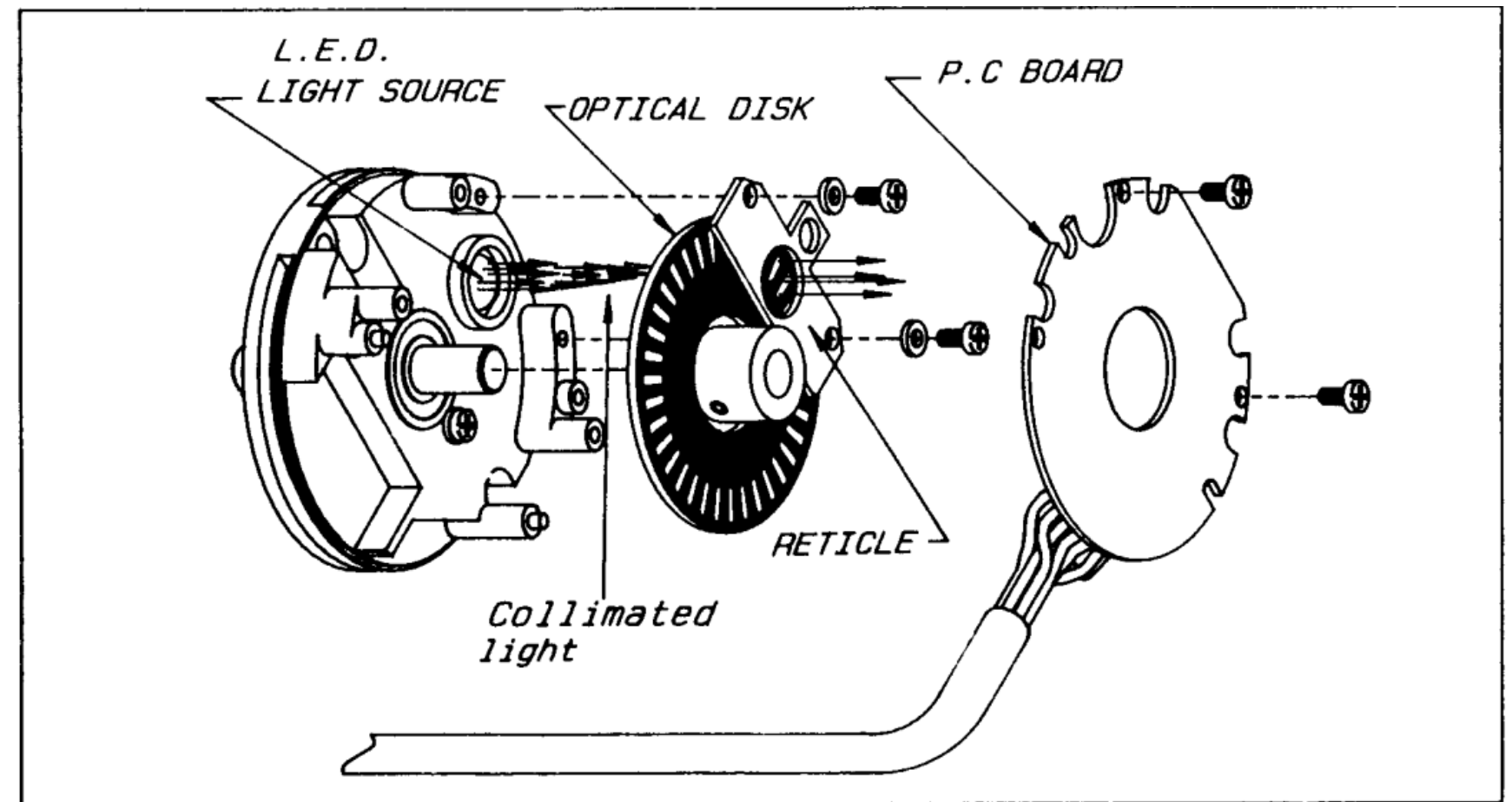
Types of encoders

- Non-Contacting
 - Capacitive
 - High resolution
 - More robust against environment
 - Susceptible to noise and interference
 - Magnetic
 - Lower resolution
 - More robust against environmental conditions
 - Watch out for magnetic interference (problem with stepper motors)



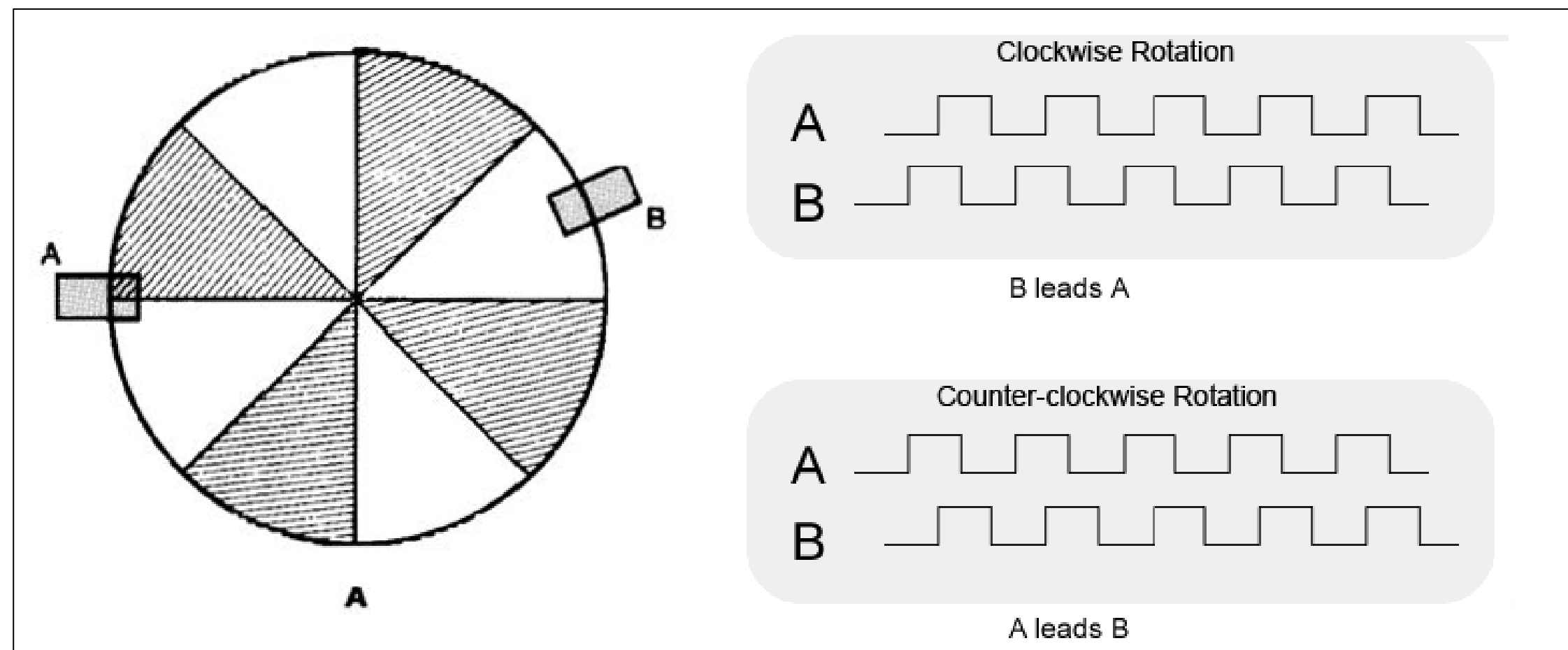
Optical encoders

- Anatomy of an optical encoder
 - Emitter
 - Detector
 - Coded disk



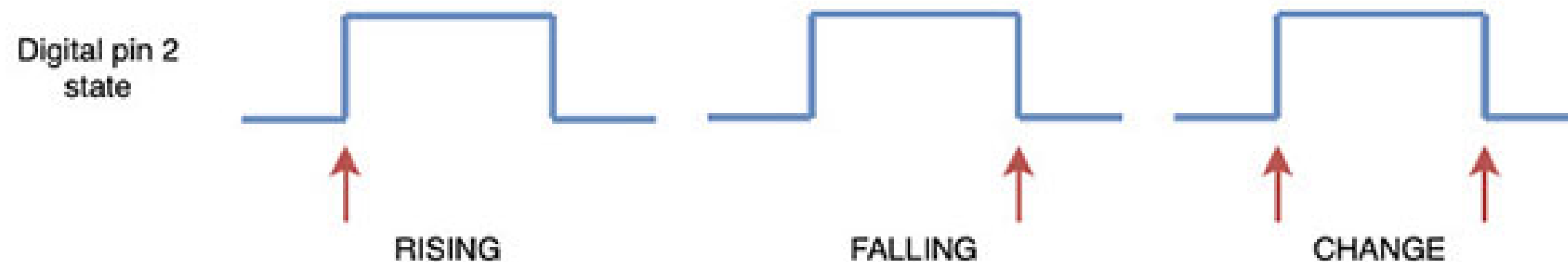
Quadrature encoders

- Counts per line (4)
- Detecting the direction of the motion is based on the offset of the two signals



Interrupts

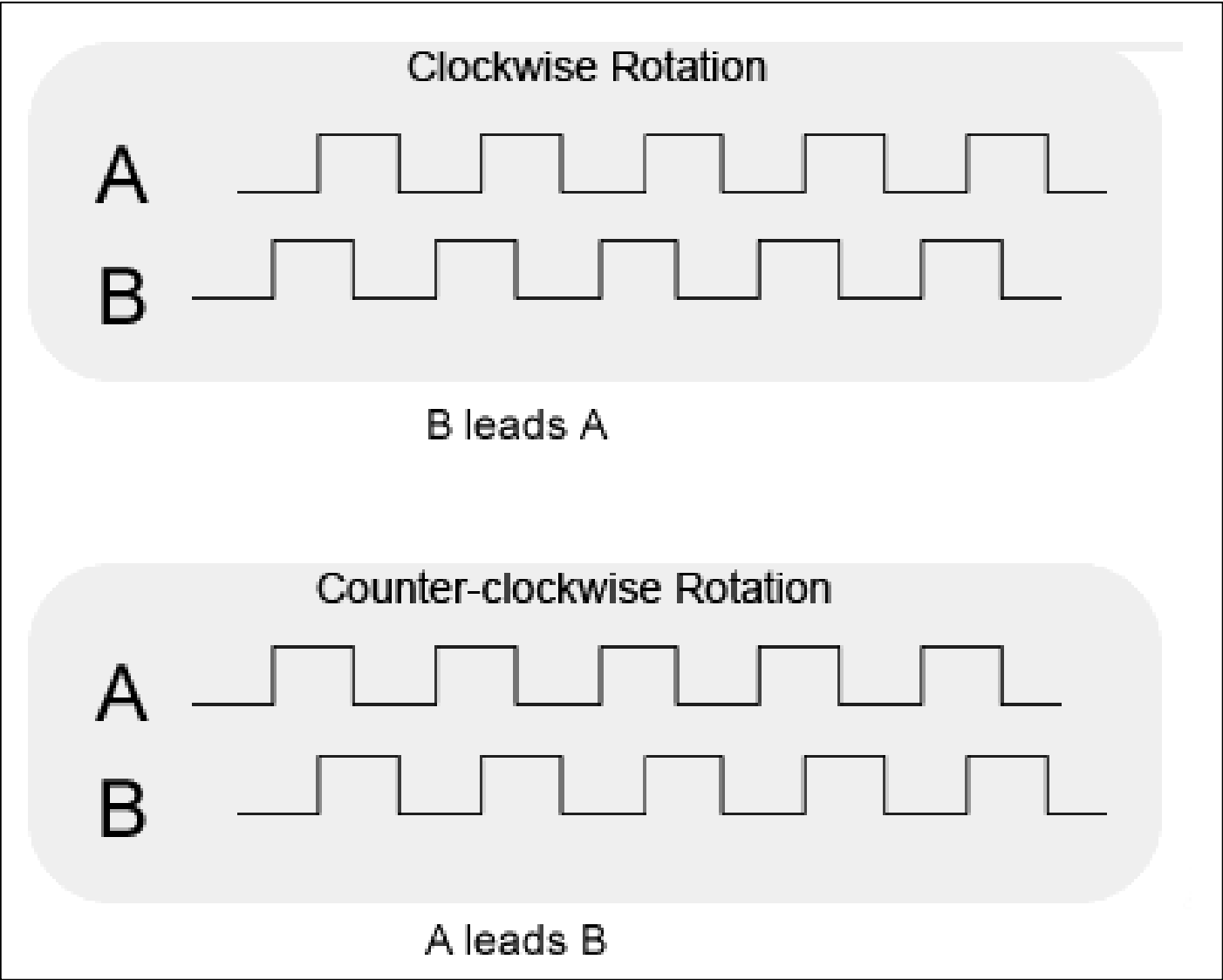
- Cause your code to run a function *immediately* when triggered
 - Hardware interrupts - Arduino UNO: pins 2, 3
 - Software interrupts
 - Timer interrupts
- Interrupts the execution of any other code.



Quadrature encoders

```
//
//          Pin1 _____ Pin1
// negative <---          ---> positive
//          Pin2 _____ Pin2

//      new      new      old      old
//      pin2     pin1     pin2     pin1      Result
//      ----     ----     ----     ----      -
//      0         0        0         0        no movement
//      0         0        0         1         +1
//      0         0        1         0         -1
//      0         0        1         1         +2 (assume pin1 edges only)
//      0         1        0         0         -1
//      0         1        0         1        no movement
//      0         1        1         0         -2 (assume pin1 edges only)
//      0         1        1         1         +1
//      1         0        0         0         +1
//      1         0        0         1         -2 (assume pin1 edges only)
//      1         0        1         0        no movement
//      1         0        1         1         -1
//      1         1        0         0         +2 (assume pin1 edges only)
//      1         1        0         1         -1
//      1         1        1         0         +1
//      1         1        1         1        no movement
```

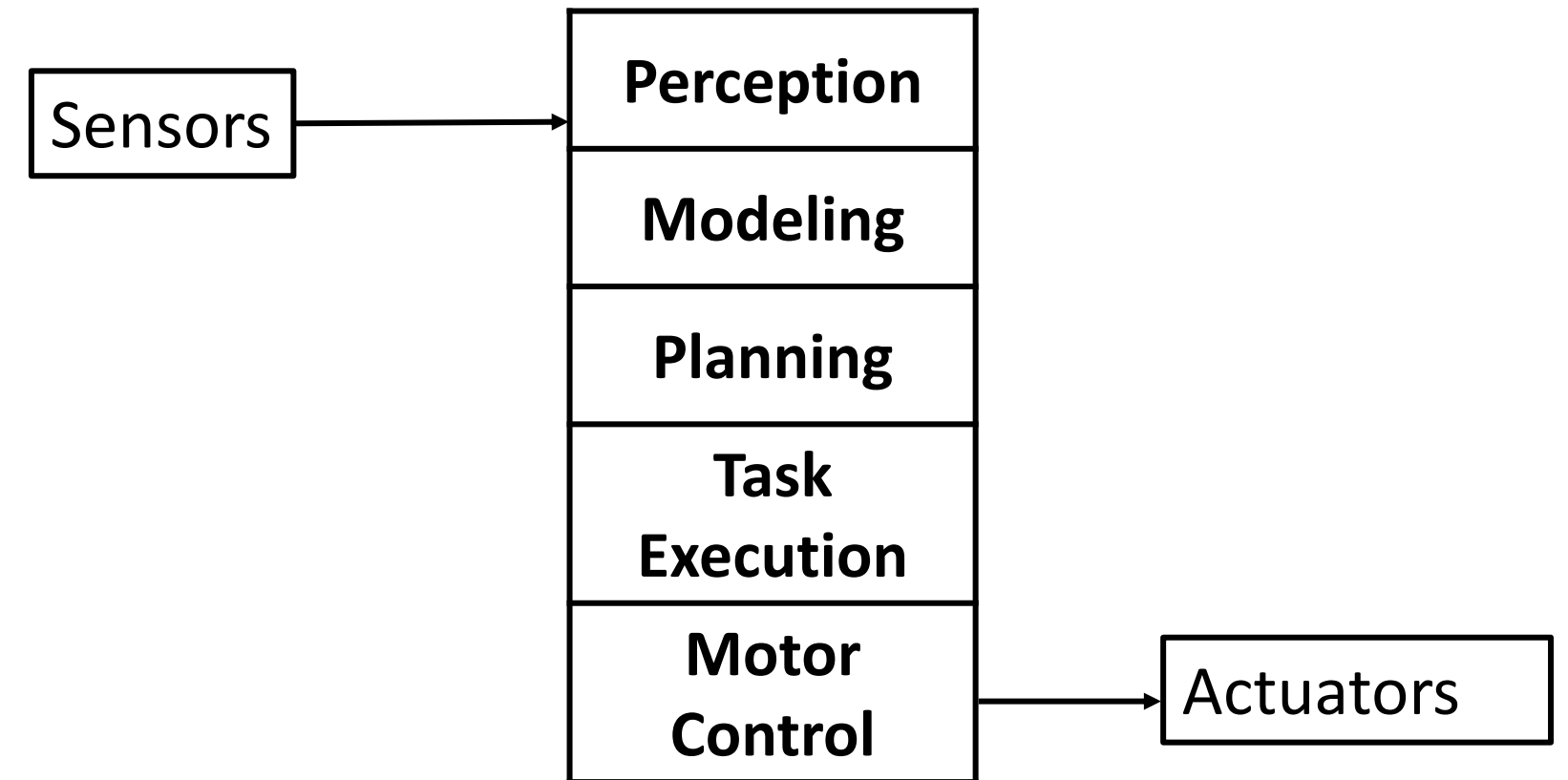


Why don't Roombas drive down stairs?
(Most of the time)



Deliberative Control

- Based on an internal representation of the world the robot interacts with
- Heavy cost in modeling the world and calculating desired output
- Model needs to be accurate to produce an accurate response (high fidelity)



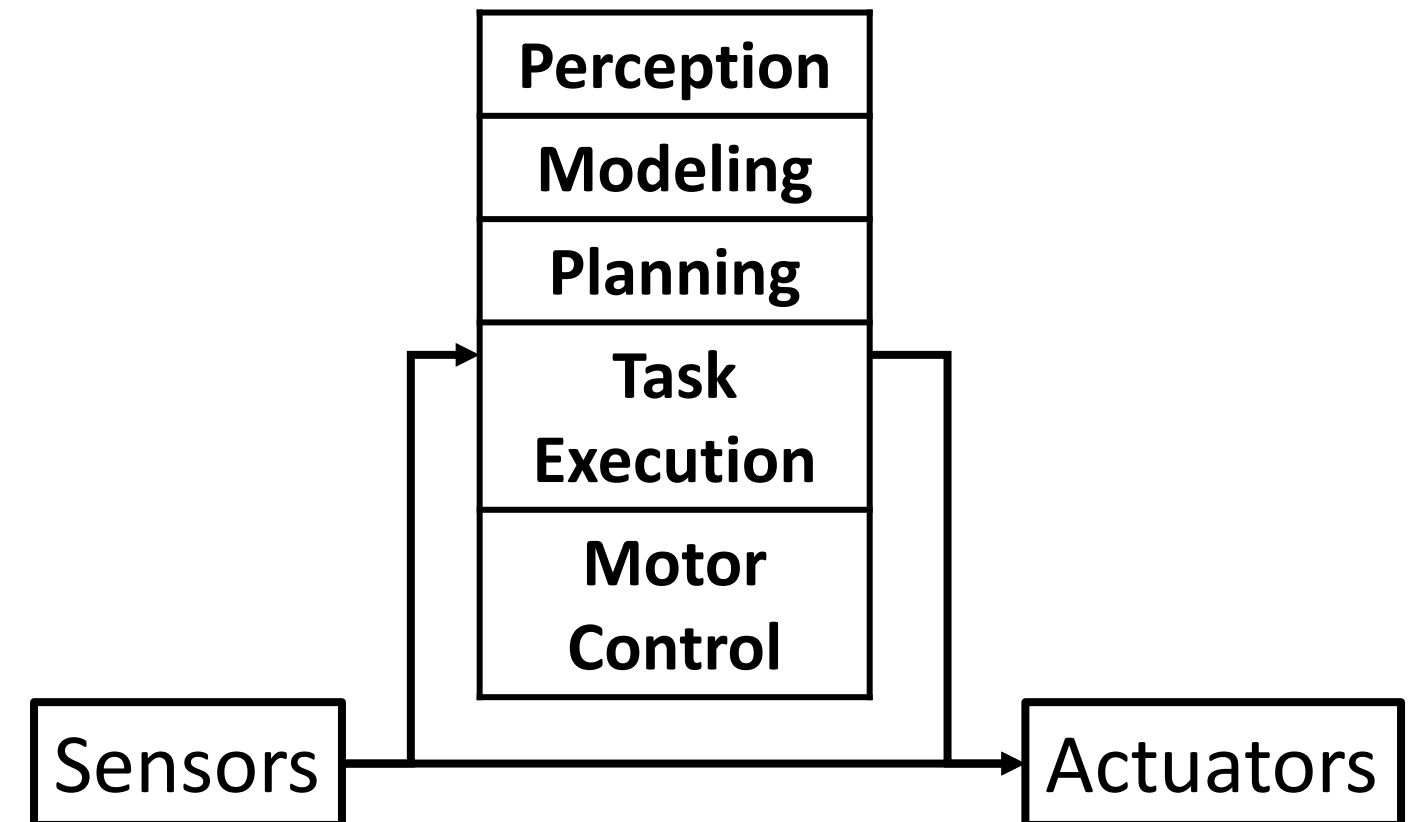
Reactive Control

- Removes the “planning” steps
- Sensor measurements directly relate to a specific action or control value
- Quick turn-around from measurement to response



Hybrid Control

- Reactive + Deliberative
- Have a higher-level control that determines whether the response should come from modeling/planning or a reaction.
- Useful for incorporating safety “reflexes” amidst higher-level thinking.



Behavior-Based Control

- Network of reactive-like control modules
- Each module represents a behavior
- Uses hierarchical levels of control
- Priorities determine which reaction take precedence

